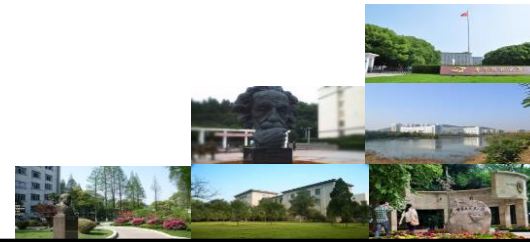
人工智能与自动化学院

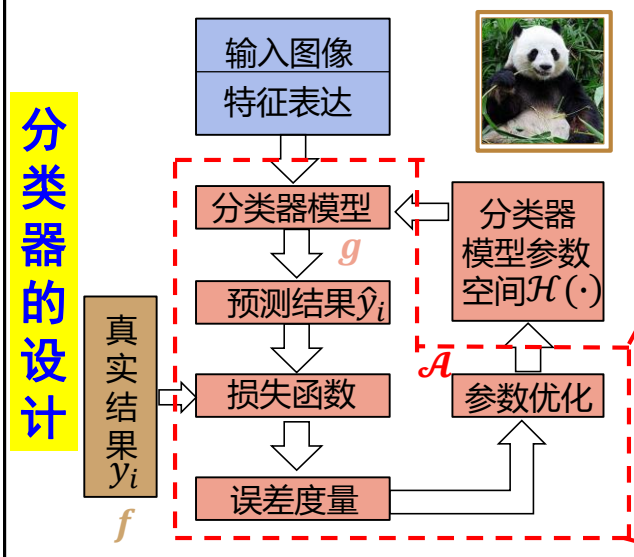
模式识别与机器学习

授课教师：杨卫东　胡静



1

1.2 模式识别方法与机器学习



算法 $\mathcal{A}$

① 感知器
② 线性回归
③ Fisher线性判别
④ 逻辑斯蒂回归
⑤ 非线性变换
⑥ 线性支撑向量机
⑦ 对偶SVM与核SVM
⑧ 多类分类
⑨ 神经网络与深度学习
⑩ 卷积神经网络
。。。

人工智能与自动化学院　2

第四讲　Fisher线性判别 (*Fisher Discriminant Analysis*)

4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)

4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)

4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

3.1 线性回归问题

为什么需要回归?

分类(*Classification*)



输出结果(*Output*):
类别标签(Class label)

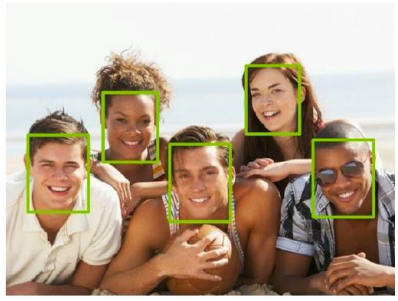
检测(分类+定位)
Detection(Classification+ Localization)



输出结果(*Output*): 类别标签+位置坐标
Class label + Box in the image (x, y, w, h)

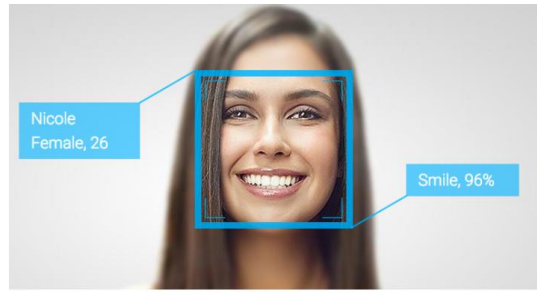
4.1 Fisher线性判别动机

应用示例:



人脸检测

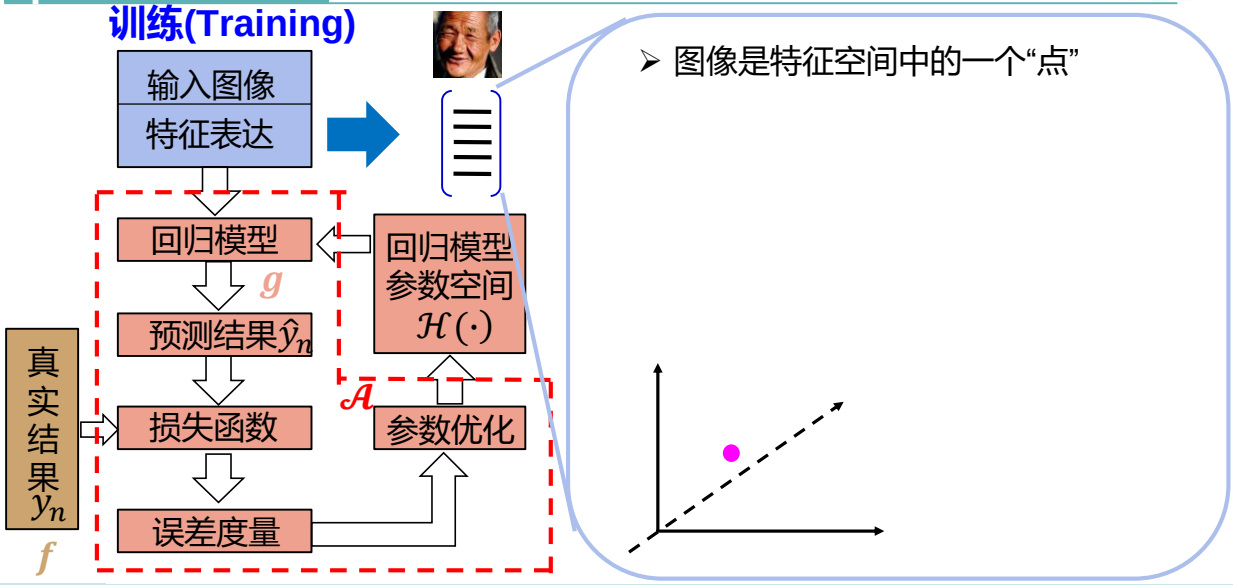
(Detection finds the faces in images)



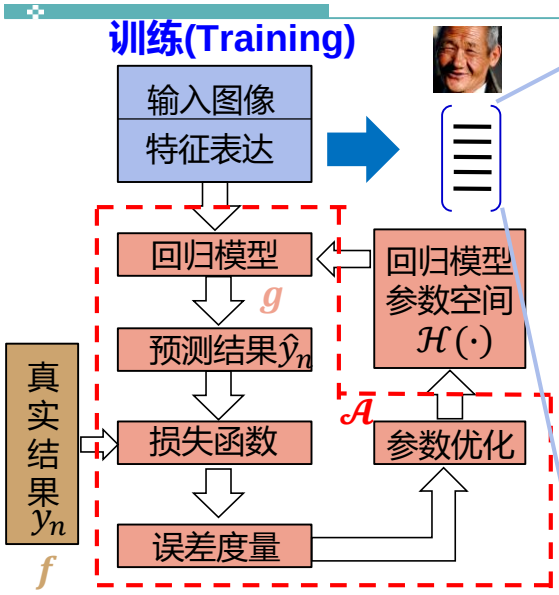
人脸识别

(Recognition recognizes WHO the person is)

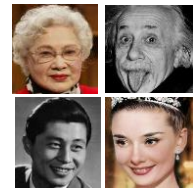
4.1 Fisher线性判别动机

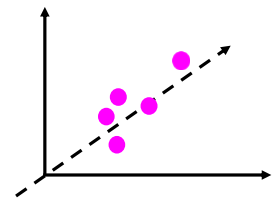


4.1 Fisher线性判别动机

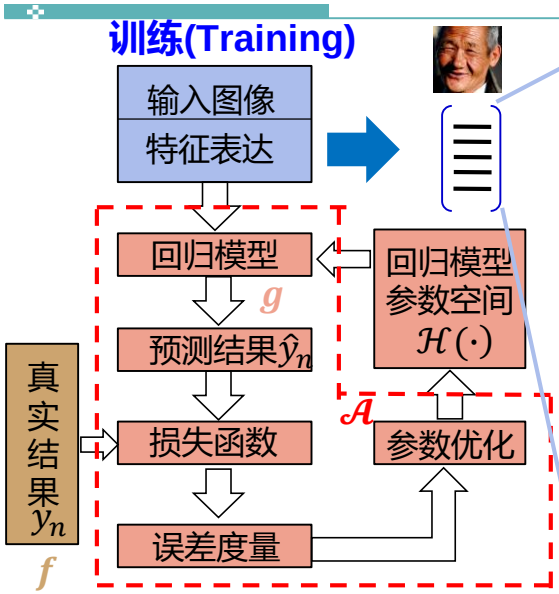


- 图像是特征空间中的一个“点”

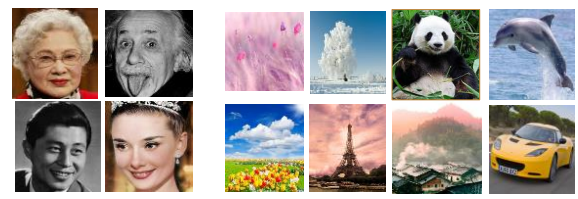


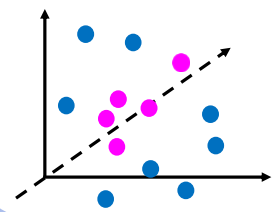


4.1 Fisher线性判别动机

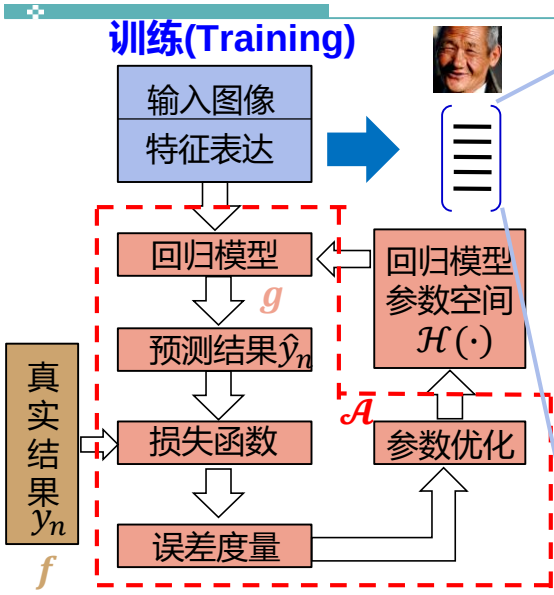


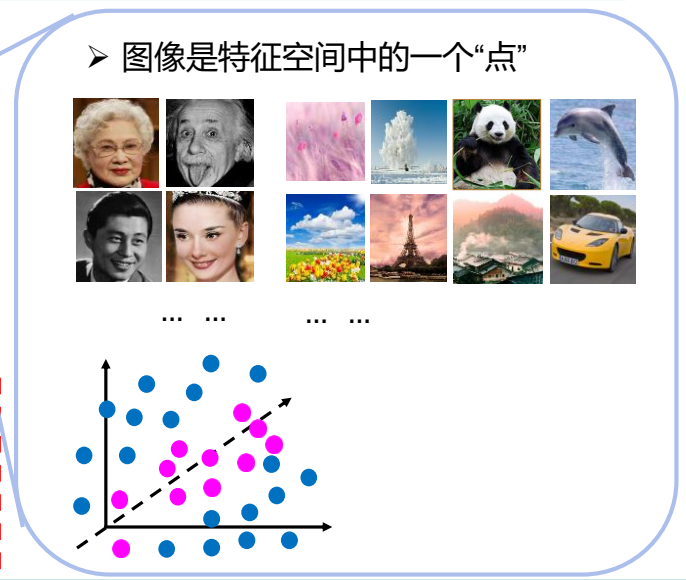
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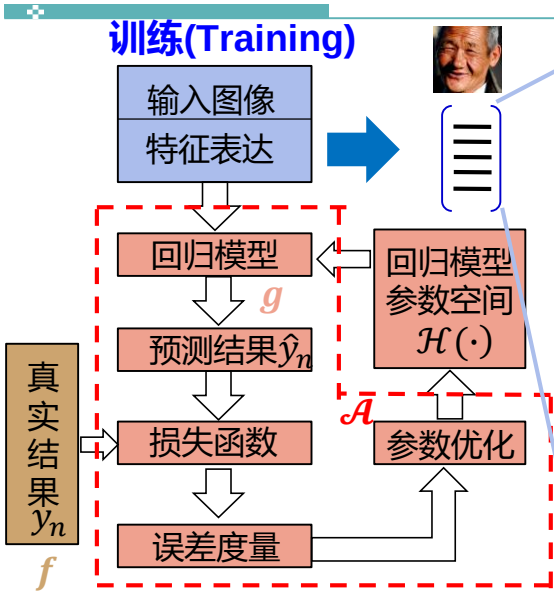


4.1 Fisher线性判别动机



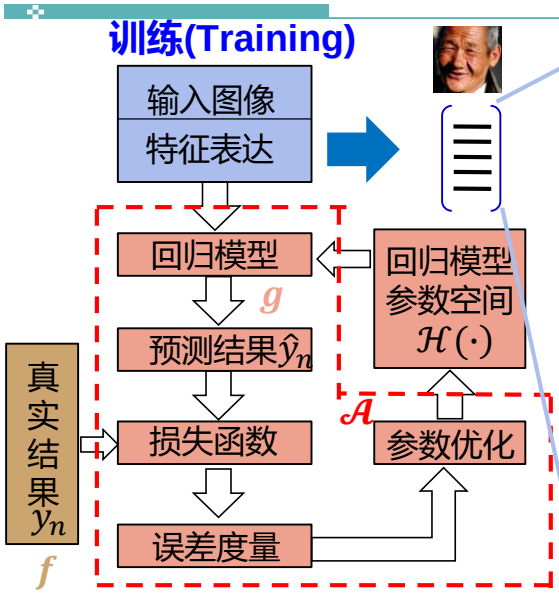


4.1 Fisher线性判别动机





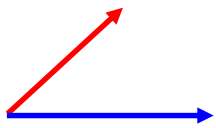
4.1 Fisher线性判别动机





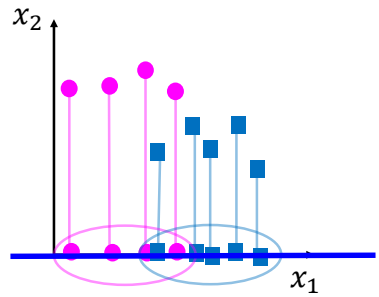
4.1 Fisher线性判别动机

投影表示：



$w^T\mathbf{x} = \|\mathbf{x}\|\|w\|cos\theta$

将 $\mathbf{x}$ 向 w 投影
（Project $\mathbf{x}$to$w$）

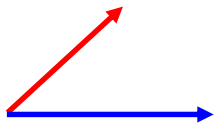


不是好的投影
（*Poor Projection*）

$g(\mathbf{x}) = w^T\mathbf{x}$

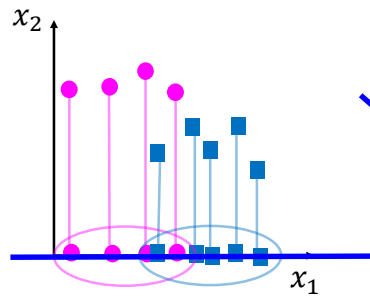
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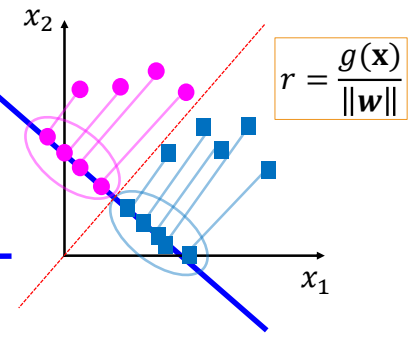


$w^T x = \|x\|\|w\|cos\theta$

将 x 向 w 投影
（Project xtow）



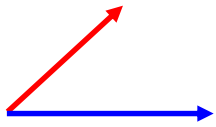
不是好的投影
（*Poor Projection*）



好的投影
（*Good Projection*）

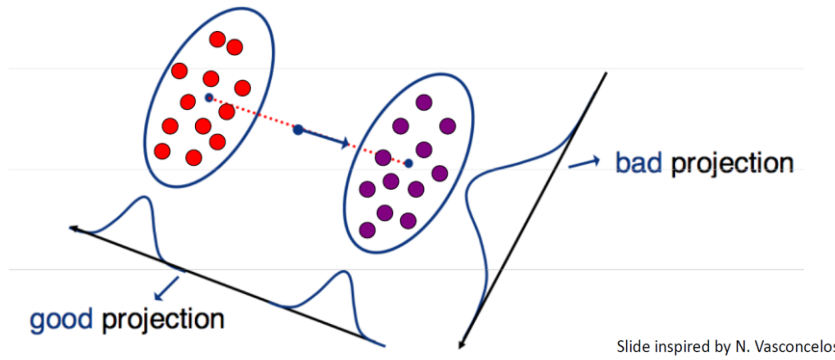
4.1 Fisher线性判别动机

投影表示:



$w^T x = \|x\|\|w\|cos\theta$

将 x 向 w 投影
（Project xtow）

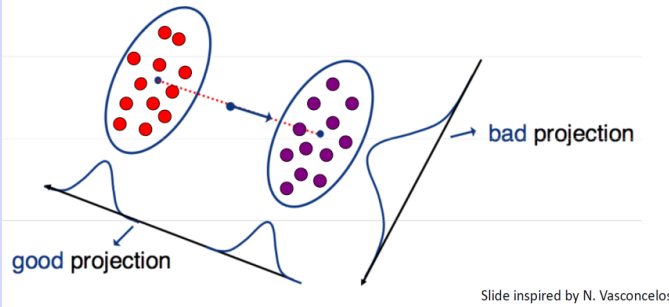


Fisher线性判别的目的：在两个类别之间找到最好的区分
（*find the best separation between two classes*）

4.1 Fisher线性判别动机

Fisher线性判别的目的：

- 在尽可能保留类别可区分性的前提下实现降维
  （*Perform dimensionality reduction "while preserving as much of the class discriminatory information as possible".*）
- 找到让类别最好区分的投影方向
  （*Seeks to find directions along which the classes are best separated.*）
- 同时考虑类内散布和类间散布
  （*Takes into consideration the scatter within-classes but also the scatter between-classes.*）

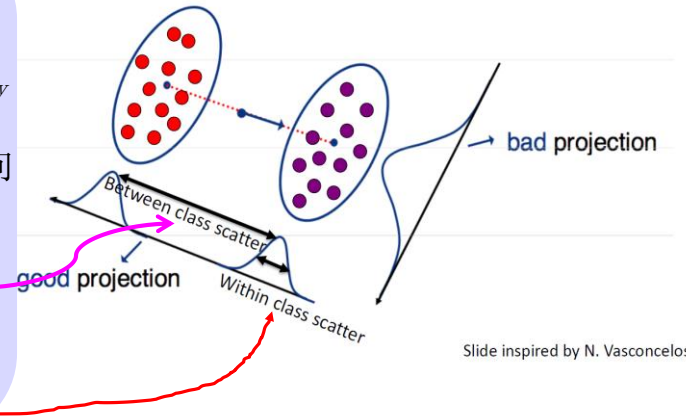


Slide inspired by N. Vasconcelos

4.1 Fisher线性判别动机

Fisher线性判别的目的：

- 在尽可能保留类别可区分性的前提下实现维数减少
  （*Perform dimensionality reduction "while preserving as much of the class discriminatory information as possible".*）
- 找到让类别最好区分的投影方向
  （*Seeks to find directions along which the classes are best separated.*）
- 同时考虑类内散布和类间散布
  （*Takes into consideration the scatter within-classes but also the scatter between-classes.*）



Slide inspired by N. Vasconcelos

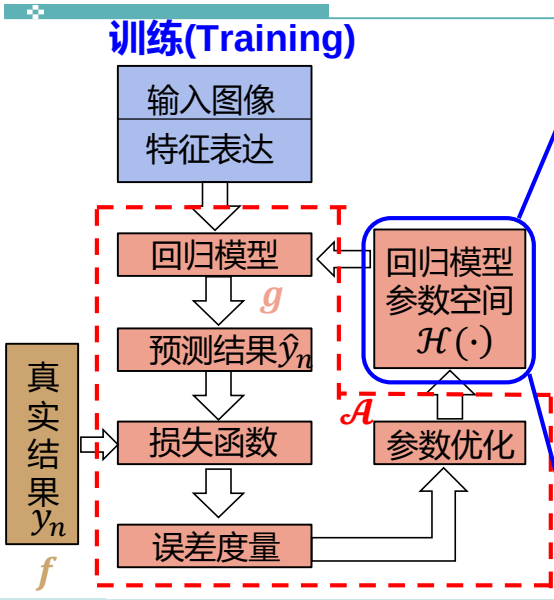
第四讲　Fisher线性判别 (*Fisher Discriminant Analysis*)

4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)

4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)

4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

3.1 线性回归问题



$\mathcal{H}(\cdot)$ 像什么?

输入： $\mathbf{x} = (x_1, x_2, \ldots x_d)^T$

计算输入 $\mathbf{x}$ 的加权和： $\hat{y} = \sum_{i=1}^{d} w_i x_i$

回归模型的参数空间： $h(\quad) \in \mathcal{H}(\cdot)$

$h(\mathbf{x}) = \mathbf{w}^T\mathbf{x}$

线性回归的目的：找到误差最小的拟合模型

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

学习最佳投影$\boldsymbol{w}^*$，它能将所有样本投影到$\boldsymbol{w}^*$的方向

假设 $s = \boldsymbol{w}^T\mathbf{x}$ $\quad \mathbf{x} \in \mathcal{R}^d, \quad s \in \mathcal{R}^1$

类别集合： $\mathcal{C} = \{c|(1,-1)\}$

第 c 个类别的均值为： $\boldsymbol{\mu}_c = E[\mathbf{x}|y=c] = \frac{1}{N_c}\sum_{n=1}^{N_c}[\mathbf{x}_n|y=c]$

第 c 个类别的协方差为： $\boldsymbol{\Sigma}_c = E[(\mathbf{x}-\mu_c)(\mathbf{x}-\mu_c)^T|y=c]$

$= \sum_{n=1}^{N_c}[(\mathbf{x}_n-\mu_c)(\mathbf{x}_n-\mu_c)^T|y=c]$

4.2 Fisher线性判别分析

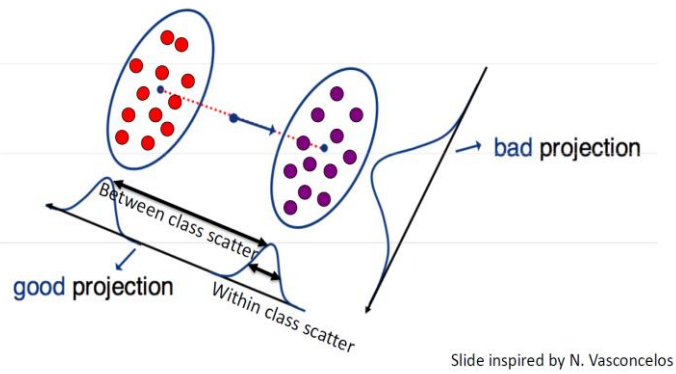
二分类问题的Fisher线性判别：

学习最佳投影$\boldsymbol{w}^*$的目标函数：

$$J(\boldsymbol{w}) = \frac{between\ class\ scatter}{within\ class\ scatter}$$

$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{argmax}\, J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = \frac{(E[s|y=1]-E[s|y=-1])^2}{var[s|y=1]+var[s|y=-1]}$$



Slide inspired by N. Vasconcelos

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

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$$J(\boldsymbol{w}) = \frac{(E[s|y=1]-E[s|y=-1])^2}{var[s|y=1]+var[s|y=-1]}$$

$$\begin{aligned}
&(E[s|y=1]-E[s|y=-1])^2 \\
&= (E[\boldsymbol{w}^T\mathbf{x}|y=1]-E[\boldsymbol{w}^T\mathbf{x}|y=-1])^2 \\
&= (\boldsymbol{w}^T(E[\mathbf{x}|y=1]-E[\mathbf{x}|y=-1]))^2 \\
&= (\boldsymbol{w}^T(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1}))^2 \\
&= \boldsymbol{w}^T(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})\,(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})^T\boldsymbol{w}
\end{aligned}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

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$$(E[s|y=1]-E[s|y=-1])^2 = \boldsymbol{w}^T(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})\,(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})^T\boldsymbol{w}$$

$$\begin{aligned}
var[s|y=c] &= E[(s-E[s|y=c])^2] \\
&= E[(\boldsymbol{w}^T\mathbf{x}-E[\boldsymbol{w}^T\mathbf{x}|y=c])^2] \\
&= E[(\boldsymbol{w}^T(\mathbf{x}-E[\mathbf{x}|y=c]))^2] \\
&= E[(\boldsymbol{w}^T(\mathbf{x}-\boldsymbol{\mu}_c))^2] \\
&= E[\boldsymbol{w}^T(\mathbf{x}-\boldsymbol{\mu}_c)(\mathbf{x}-\boldsymbol{\mu}_c)^T\boldsymbol{w}] \\
&= \boldsymbol{w}^T E[(\mathbf{x}-\boldsymbol{\mu}_c)(\mathbf{x}-\boldsymbol{\mu}_c)^T]\boldsymbol{w} \\
&= \boldsymbol{w}^T\boldsymbol{\Sigma}_c\boldsymbol{w}
\end{aligned}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

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$$J(\boldsymbol{w}) = \frac{(E[s|y=1]-E[s|y=-1])^2}{var[s|y=1]+var[s|y=-1]}$$

$$(E[s|y=1]-E[s|y=-1])^2 = \boldsymbol{w}^T(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})^T\boldsymbol{w}$$

$$var[s|y=c] = E[(s-E[s|y=c])^2] = \boldsymbol{w}^T\boldsymbol{\Sigma}_c\boldsymbol{w}$$

4.2 Fisher线性判别分析

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$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})(\boldsymbol{\mu}_1-\boldsymbol{\mu}_{-1})^T\boldsymbol{w}}{\boldsymbol{w}^T\boldsymbol{\Sigma}_1\boldsymbol{w}+\boldsymbol{w}^T\boldsymbol{\Sigma}_{-1}\boldsymbol{w}}$$

4.2 Fisher线性判别分析

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$$\boldsymbol{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})\,(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T$$

$$\boldsymbol{S}_w = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

学习最佳投影$\boldsymbol{w}^*$的目标函数：

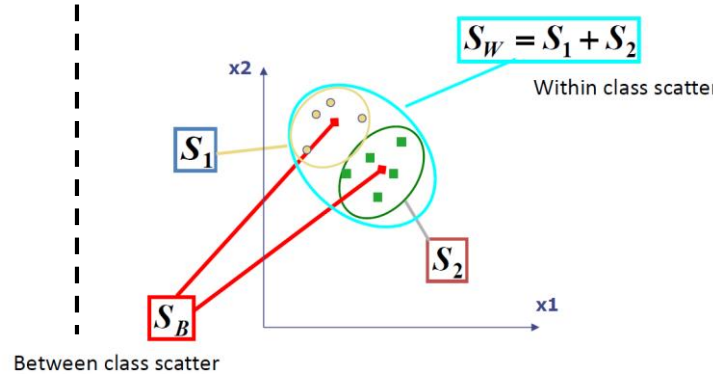
$$J(\boldsymbol{w}) = \frac{between\ class\ scatter}{within\ class\ scatter}$$

$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{argmax}\, J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T\boldsymbol{S}_B\boldsymbol{w}}{\boldsymbol{w}^T\boldsymbol{S}_w\boldsymbol{w}}$$

$$\boldsymbol{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})\,(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T$$

$$\boldsymbol{S}_w = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1} = \boldsymbol{S}_1 + \boldsymbol{S}_2$$



4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

学习最佳投影 $\boldsymbol{w}^*$ 的目标函数：

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$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{argmax}\, J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w}}$$

最大化目标函数问题转化为：

$$\max_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} \quad Subject\ to\ \boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w} = \boldsymbol{K}$$

利用拉格朗日乘子法(*Lagrange multipliers*)：

$$L(\boldsymbol{w}, \lambda) = \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} + \lambda(\boldsymbol{K} - \boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w})$$

$$= \boldsymbol{w}^T (\boldsymbol{S}_B - \lambda \boldsymbol{S}_w) \boldsymbol{w} + \lambda \boldsymbol{K}$$

$$\nabla L_w(\boldsymbol{w}, \lambda) = \frac{\partial L(\boldsymbol{w}, \lambda)}{\partial \boldsymbol{w}} = \boldsymbol{0}^T$$

$$2(\boldsymbol{S}_B - \lambda \boldsymbol{S}_w)\boldsymbol{w} = \boldsymbol{0} \Rightarrow \boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{S}_w \boldsymbol{w}$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

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$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{argmax}\, J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w}}$$

$$\max_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} \quad Subject\ to\ \boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w} = \boldsymbol{K}$$

$$L(\boldsymbol{w}, \lambda) = \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} + \lambda(\boldsymbol{K} - \boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w})$$

$$\nabla L_w(\boldsymbol{w}, \lambda) = \boldsymbol{0}$$

$$\boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{S}_w \boldsymbol{w}$$

如果$\boldsymbol{S}_w^{-1} = (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1})^{-1}$ 存在, 则有：

$$\boldsymbol{S}_w^{-1} \boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{w}$$

$$\boldsymbol{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \boldsymbol{w} = \lambda \boldsymbol{w}$$

$$\boldsymbol{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})\, a = \lambda \boldsymbol{w}$$ $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})^T \boldsymbol{w}$=$a$标量

$$\boldsymbol{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}) = \frac{\lambda}{a} \boldsymbol{w}$$

只关注投影向量的方向：

$$\boldsymbol{w}^* = \boldsymbol{S}_w^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})$$

4.2 Fisher线性判别分析

二分类问题的Fisher线性判别：

学习最佳投影 $\boldsymbol{w}^*$ 的目标函数：

$$J(\boldsymbol{w}) = \frac{between\ class\ scatter}{within\ class\ scatter}$$

$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{argmax}\, J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w}}$$

$$\max_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} \quad Subject\ to\ \boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w} = \boldsymbol{K}$$

$$L(\boldsymbol{w}, \lambda) = \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w} - \lambda(\boldsymbol{w}^T \boldsymbol{S}_w \boldsymbol{w} - \boldsymbol{K})$$

$$\nabla L_w(\boldsymbol{w}, \lambda) = \boldsymbol{0} \Rightarrow \boldsymbol{w}^* = \boldsymbol{S}_w^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})$$

找到投影向量后，对任一测试样本 $\mathbf{x}$ ：

$$s = \boldsymbol{w}^{*T}\mathbf{x} = (\boldsymbol{S}_w^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}))^T\mathbf{x}$$

假设类别的判别门限设为 s' ：

$$s' = \frac{\boldsymbol{w}^{*T}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_{-1})}{2}$$

对任一测试样本 $\mathbf{x}$ 所属类别的判断：

$$\begin{cases} y = 1 & if\ s = \boldsymbol{w}^{*T}\mathbf{x} > s' \\ y = -1 & if\ s = \boldsymbol{w}^{*T}\mathbf{x} < s' \end{cases}$$

第四讲 Fisher线性判别 (*Fisher Discriminant Analysis*)

4.1 Fisher线性判别动机 (*The goal of Fisher Linear Discriminant*)

4.2 Fisher线性判别分析 (*Fisher Discriminant Analysis*)

4.3 Fisher线性判别算法 (*Fisher Discriminant Algorithm*)

4.3 Fisher线性判别算法

二分类问题的Fisher线性判别算法：

① 获取具有标签的两类样本

② 依据下式得到$\boldsymbol{\mu}_1$和$\boldsymbol{\mu}_{-1}$：

$$\boldsymbol{\mu}_c = \frac{1}{N_c}\sum_{n=1}^{N_c}[\mathbf{x}_n|y=c]$$

③ 依据下式得到$\boldsymbol{\Sigma}_1$和$\boldsymbol{\Sigma}_{-1}$：

$$\boldsymbol{\Sigma}_c = \sum_{n=1}^{N_c}[(\mathbf{x}_n-\mu_c)(\mathbf{x}_n-\mu_c)^T|y=c]$$

④ 计算**类内总离差阵**：$\boldsymbol{S}_w = \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_{-1}$

⑤ 计算类内总离差阵的逆：$\boldsymbol{S}_w^{-1}$

⑥ 计算最佳投影：$\boldsymbol{w}^* = \boldsymbol{S}_w^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1})$

⑦ 计算判别门限$\boldsymbol{s}'$：$s' = \frac{\boldsymbol{w}^{*T}(\boldsymbol{\mu}_1+\boldsymbol{\mu}_{-1})}{2}$

⑧ 对任一测试样本 $\mathbf{x}$ ：

$$\begin{cases} y=1 & if\ s=\boldsymbol{w}^{*T}\mathbf{x} > s' \\ y=-1 & if\ s=\boldsymbol{w}^{*T}\mathbf{x} < s' \end{cases}$$

第四讲　Fisher线性判别 (*Fisher Discriminant Analysis*)

4.1 Fisher线性判别动机

在尽可能保留类别可区分性的前提下实现维数减少

4.2 Fisher线性判别分析

找到让类别最好区分的投影方向

4.3 Fisher线性判别算法

通过计算类内散布和类间散布，找到最佳 $\boldsymbol{w}^*$和判别门限s'